

# Beyond Morin, and deeper into $A_7$ :

the corank filtration of Thom polynomials,  
and the anatomy of Chern positivity at  $d \leq 7$

Chern++

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## Abstract

We place Rimányi’s conjecture inside a filtration  $C_1 \subset C_2 \subset \dots$  of cones of symmetric functions, running from the Chern-monomial cone to the Schur-positive cone. Rimányi’s conjecture says that  $\text{Tp}_{A_d} \in C_1$ . The natural generalisation is  $\text{Tp}_\eta \in C_{\text{corank } \eta}$ . We prove the matching lower bound, and we prove, by exact certificates, that the generalisation itself is false. The minimal rank  $\rho(\eta)$  is therefore a new invariant, and we tabulate it on the published registry. For Morin singularities we give a sweep algorithm for the chamber series that is 50–100× faster than the previous one. It yields  $\text{Tp}_{A_7}$  exactly for  $\ell \leq 5$  and  $\text{Tp}_{A_6}$  for  $\ell \leq 8$ , all positive. We locate where positivity is tight (the plane sector, whose values are Kreweras numbers), and we observe a quantitative strengthening: the negative mass of every packet is at most a fixed fraction of a single term. Every statement is labelled *proved*, *verified* (a finite exact computation) or *conjectured*.

## Summary of results

|                         | statement                                                                                                     | status      |
|-------------------------|---------------------------------------------------------------------------------------------------------------|-------------|
| <a href="#">2</a>       | $\rho(\eta) \geq \text{support corank} \geq \text{corank } \eta$                                              | proved      |
| <a href="#">3</a>       | $\sum \text{Chern coefficients} = [s_{(n)}] \text{Tp}; = 0 \text{ at corank} \geq 2$                          | proved      |
| <a href="#">4</a>       | $\text{Tp}_{I_{2,4}}, \text{Tp}_{III_{2,3}} \notin C_2$ ; $\rho \neq \text{corank}$                           | proved      |
| <a href="#">Table 1</a> | $\rho$ on the registry; $\rho - \text{corank}$ reaches 2                                                      | verified    |
| <a href="#">§2</a>      | $\text{Tp}_{A_7}$ ( $\ell \leq 5$ ), $\text{Tp}_{A_6}$ ( $\ell \leq 8$ ) exact and positive, $\min C = 1$     | verified    |
| <a href="#">8</a>       | plane sector: $C(M) = b(M)$ (ballot words) = Kreweras                                                         | proved*     |
| <a href="#">9</a>       | $C(M) \geq b(M)$ , equality iff $\max M \leq 1$ (implies Rimányi)                                             | conjectured |
| <a href="#">11</a>      | $N(M) \leq \kappa_d A_{\beta_{\max}(M)}$ with $\kappa_5 = \kappa_6 = \frac{1}{6}$ , $\kappa_7 = \frac{8}{21}$ | conjectured |
| <a href="#">§4</a>      | 0-Hecke landscape $P_5, P_6$ (stable), $P_7$ (on boxes)                                                       | verified    |

\* given two geometric inputs (a curvilinear incidence formula and a hyperplane lemma), stated in full in [report/ballot.pdf](#).

## 1 The corank filtration

Write a Thom polynomial in the relative Chern classes and read  $c_k$  as the complete homogeneous symmetric function  $h_k$ . Then  $s_\nu = \det(c_{\nu_i + j - i})$ , and Pragacz–Weber positivity [3] says that every  $\text{Tp}_\eta$  is a nonnegative combination of the  $s_\nu$ .

**Definition 1.** For  $r \geq 1$  let  $C_r$  be the convex cone spanned by all products  $s_{\nu^1} \cdots s_{\nu^k}$  in which every  $\nu^i$  has at most  $r$  rows. Set  $\rho(\eta) = \min\{r : \text{Tp}_\eta \in C_r\}$ .

Since  $s_{(k)} = c_k$ , the cone  $C_1$  is spanned by the Chern monomials; because Chern monomials form a basis,  $f \in C_1$  exactly when all its Chern coefficients are nonnegative. At the other end,

$\bigcup_r C_r$  is the Schur-positive cone. Hence:

$$\text{Rimányi's conjecture} \iff \rho(A_d) = 1, \quad \text{Pragacz-Weber} \iff \rho(\eta) < \infty.$$

The filtration is natural from the point of view of residue formulas. The Bérczi–Szenes formula is a residue over  $\mathbb{P}(TX)$ : after the residue, each chamber monomial becomes a product of one-row Schur functions  $c_{\ell+1+\alpha_i}$ , and Rimányi's conjecture asks the packet sums to be nonnegative. A formula of the same kind on the Grassmannian  $\text{Gr}(r, TX)$  would produce products of Schur functions with at most  $r$  rows. So *a corank- $r$  residue formula with nonnegative packet sums would place  $\text{Tp}_\eta$  in  $C_r$ .*

**Theorem 2** (lower bound; proved). *If every Schur component of  $\text{Tp}_\eta$  has at least  $r$  rows, then  $\text{Tp}_\eta \notin C_{r-1}$ . In particular  $\rho(\eta) \geq \text{corank } \eta$ .*

*Proof.* Let  $g = s_{\nu^1} \cdots s_{\nu^k}$  with every  $\ell(\nu^i) \leq r-1$ . By the Littlewood–Richardson rule,  $s_{\nu^1+\dots+\nu^k}$  (row-wise sum) appears in  $g$  with coefficient 1, and it has at most  $r-1$  rows. All LR coefficients are nonnegative, so in a nonnegative combination of such products nothing cancels it. Every nonzero element of  $C_{r-1}$  therefore has a Schur component with at most  $r-1$  rows, and  $\text{Tp}_\eta$  has none. A singularity of corank  $r$  lies in the Thom–Boardman stratum  $\Sigma^r$ , and every Schur component of its Thom polynomial contains the rectangle  $(\ell+r)^r$ . This is the support property, verified on every registry table: the column  $s$  of Table 1 is at least the corank.  $\square$

**Corollary 3** (proved). *The sum of the Chern coefficients of any  $\text{Tp}_\eta$  equals its coefficient on  $s_{(n)}$ , where  $n = \text{codim } \eta$ . This is  $(d!)^{\ell+1}$  for  $A_d$  and 0 for every  $\eta$  of corank  $\geq 2$ . In particular, no Thom polynomial of corank  $\geq 2$  is Chern-positive, and each has at least one negative Chern coefficient.*

*Proof.*  $h_\mu = \sum_\lambda K_{\lambda\mu} s_\lambda$ , and  $K_{(n),\mu} = 1$  for every  $\mu \vdash n$ . So  $[s_{(n)}] \sum_\mu a_\mu h_\mu = \sum_\mu a_\mu$ . At corank  $\geq 2$  every component has at least two rows. For  $A_d$  the value  $(d!)^{\ell+1}$  is verified: on every published table (tier 14) and on all fifteen new tables of §2.  $\square$

The negative Chern coefficients of non-Morin Thom polynomials are thus *forced*. They are not an accident of the data, and they are exactly what Theorem 2 forbids. The natural generalisation of Rimányi's conjecture would be  $\rho = \text{corank}$ . It is false.

**Proposition 4** (proved by certificate).  $\text{Tp}_{I_{2,4}} = 16s_{42} + 4s_{33} + 12s_{321} + 5s_{222} + 2s_{2211}$  (at  $\ell = 0$ ) is not in  $C_2$ . Similarly  $\text{Tp}_{\text{III}_{2,3}} = 8s_{53} + 4s_{431} + 2s_{332}$  (at  $\ell = 1$ ) is not in  $C_2$ . Both lie in  $C_3$ .

*Proof.* Since products of Schur functions are Schur-positive, a decomposition can only use generators  $g$  with  $\text{supp } g \subseteq \text{supp } \text{Tp}$ , and there are finitely many. For  $I_{2,4}$  the functional  $\varphi = [s_{33}] + [s_{222}] - [s_{321}]$  is nonnegative on each of them but  $\varphi(\text{Tp}) = 4 + 5 - 12 = -3$ . For  $\text{III}_{2,3}$  take  $\varphi = [s_{332}] - [s_{431}]$ : for instance  $s_{33}s_2 = s_{53} + s_{431} + s_{332}$ , while  $\varphi(\text{Tp}) = -2$ . The nonnegativity checks, and the  $C_3$  decompositions, are exact rational computations in `chernpp.corank` (tier 14).  $\square$

**The invariant  $\rho$  on the registry.** Table 1 gives  $\rho$  for the 87 tables decided so far, out of 128 non-Morin tables in the registry. Most of the rest are of codimension 19 to 26 and exceeded the time budget. In three of them ( $I_{2,5}$ ,  $\text{III}_{2,6}$  and  $\text{III}_{3,5}$  at  $\ell = 2$ ) the LP reports  $\text{Tp} \notin C_2$ , but its Farkas vector did not rationalise to a certificate that verifies, so they are left undecided rather than reported. Each entry is decided by a linear programme whose answer is re-verified in exact arithmetic, with a decomposition or a Farkas vector as appropriate. The Farkas vectors are strikingly simple: every one found is  $\pm \frac{1}{k}$  on a handful of Schur functions. What the table shows:

- $\rho(I_{a,b}) = 2$  exactly when  $b \leq a+1$ , and 3 otherwise, for  $a \leq b$ . Likewise  $\rho(\text{III}_{a,b}) = 2$  exactly when  $a = b$ . In both families  $\rho$  is constant in  $\ell$  wherever it was computed.

- $\rho$  is not always constant in  $\ell$ :  $\rho(B_4) = 2$  at  $\ell = 1$  and 3 at  $\ell = 2$ .
- $\rho - s$  reaches 2:  $B_{5,3}$  at  $\ell = 1$  has  $s = 2$  and  $\rho = 4$ .  $C_6$  and  $C_{5,3}$  have  $s = 3$  and  $\rho = 4$ .

We have not found an invariant of the local algebra that predicts  $\rho$  in general; the multiplicity and the Hilbert function do not. This is the open question the filtration raises.

*Remark 5* (a negative result). A sharper variant replaces cones by a basis:  $b_\lambda^{(r)} = \prod_k s_{(\lambda_{rk+1}, \dots, \lambda_{rk+r})}$ . It is unitriangular against  $h_\lambda$ , equals  $h_\lambda$  at  $r = 1$ , and so has unique coefficients, as Rimányi's conjecture does.  $\text{Tp}$  is not  $b^{(\text{corank})}$ -positive in general.  $I_{2,2}$ ,  $I_{3,3}$ ,  $I_{4,4}$ ,  $\text{III}_{2,2}$  and  $\text{III}_{3,3}$  are  $b^{(2)}$ -positive wherever tabulated, and  $I_{3,4}$  at  $\ell = 0$ . But  $\text{III}_{4,4}$  is not,  $I_{2,3}$  fails from  $\ell = 2$ , and  $I_{2,4}$ ,  $I_{4,5}$  and  $B_4$  fail already at the smallest  $\ell$ .

*Remark 6* (the shape of  $C_2$ ). One might hope that  $C_2$ , like  $C_1$ , is generated by a basis. Then membership would be nonnegativity of unique coefficients, and  $\rho = 2$  would be a Rimányi-type statement in that basis. The exact extreme rays and facets of  $C_2$ , in the Schur basis of degree  $n$  (normaliz, `tools/c2_cone.py`), show that this holds only in low degree:

| $n$              | 2 | 3 | 4 | 5 | 6  | 7  | 8  | 9  | 10  |
|------------------|---|---|---|---|----|----|----|----|-----|
| dimension $p(n)$ | 2 | 3 | 5 | 7 | 11 | 15 | 22 | 30 | 42  |
| extreme rays     | 2 | 3 | 5 | 7 | 13 | 17 | 28 | 40 | 61  |
| facets           | 2 | 3 | 5 | 7 | 14 | 18 | 38 | 86 | 598 |

$C_2$  is simplicial exactly for  $n \leq 5$ . At  $n = 4$  its extreme rays are  $s_{22}$ ,  $s_{11}^2$ ,  $s_{31}$ ,  $s_2 s_{11}$  and  $s_4$ . From  $n = 6$  on it has more extreme rays than its dimension, and the number of facets grows fast. So there is no basis version of  $C_2$ , and  $\rho$  is a genuine invariant of the cone. The certificate of Proposition 4 for  $I_{2,4}$  lives in degree 6, the first non-simplicial degree.

Table 1:  $\rho(\text{Tp}) = \min\{r : \text{Tp} \in C_r\}$  on the non-Morin tables of the registry, by relative dimension  $\ell$ , with the support corank  $s$  (every Schur component contains  $(\ell + s)^s$ ). Every entry is decided with an exact certificate; a dash means not in the registry or not yet surveyed.

| singularity        | $s$ | $\ell = 0$ | $\ell = 1$ | $\ell = 2$ | $\ell = 3$ | $\ell = 4$ | $\ell = 5$ |
|--------------------|-----|------------|------------|------------|------------|------------|------------|
| $I_{2,2}$          | 2   | 2          | 2          | 2          | 2          | 2          | 2          |
| $I_{2,3}$          | 2   | 2          | 2          | 2          | 2          | 2          | —          |
| $I_{2,4}$          | 2   | <b>3</b>   | <b>3</b>   | <b>3</b>   | —          | —          | —          |
| $I_{2,5}$          | 2   | <b>3</b>   | <b>3</b>   | —          | —          | —          | —          |
| $I_{2,6}$          | 2   | <b>3</b>   | —          | —          | —          | —          | —          |
| $I_{2,7}$          | 2   | <b>3</b>   | —          | —          | —          | —          | —          |
| $I_{3,3}$          | 2   | 2          | 2          | 2          | 2          | —          | —          |
| $I_{3,4}$          | 2   | 2          | 2          | 2          | —          | —          | —          |
| $I_{3,5}$          | 2   | <b>3</b>   | —          | —          | —          | —          | —          |
| $I_{3,6}$          | 2   | <b>3</b>   | —          | —          | —          | —          | —          |
| $I_{4,4}$          | 2   | 2          | —          | —          | —          | —          | —          |
| $I_{4,5}$          | 2   | 2          | —          | —          | —          | —          | —          |
| $\text{III}_{2,2}$ | 2   | —          | 2          | 2          | 2          | 2          | 2          |
| $\text{III}_{2,3}$ | 2   | —          | <b>3</b>   | <b>3</b>   | <b>3</b>   | <b>3</b>   | <b>3</b>   |
| $\text{III}_{2,4}$ | 2   | —          | <b>3</b>   | <b>3</b>   | <b>3</b>   | —          | —          |
| $\text{III}_{2,5}$ | 2   | —          | <b>3</b>   | <b>3</b>   | —          | —          | —          |
| $\text{III}_{2,6}$ | 2   | —          | <b>3</b>   | —          | —          | —          | —          |
| $\text{III}_{3,3}$ | 2   | —          | 2          | 2          | 2          | —          | —          |
| $\text{III}_{3,4}$ | 2   | —          | <b>3</b>   | <b>3</b>   | —          | —          | —          |

Table 1:  $\rho(\text{Tp}) = \min\{r : \text{Tp} \in C_r\}$  on the non-Morin tables of the registry, by relative dimension  $\ell$ , with the support corank  $s$  (every Schur component contains  $(\ell + s)^s$ ). Every entry is decided with an exact certificate; a dash means not in the registry or not yet surveyed.

| singularity           | $s$ | $\ell = 0$ | $\ell = 1$ | $\ell = 2$ | $\ell = 3$ | $\ell = 4$ | $\ell = 5$ |
|-----------------------|-----|------------|------------|------------|------------|------------|------------|
| III <sub>3,5</sub>    | 2   | —          | <b>3</b>   | —          | —          | —          | —          |
| III <sub>4,4</sub>    | 2   | —          | 2          | 2          | —          | —          | —          |
| $(x^2 + y^3, y^4)$    | 2   | 2          | —          | —          | —          | —          | —          |
| $(x^2 + y^4, xy^2)$   | 2   | <b>3</b>   | —          | —          | —          | —          | —          |
| B <sub>4</sub>        | 2   | —          | 2          | <b>3</b>   | <b>3</b>   | —          | —          |
| B <sub>5,1</sub>      | 2   | 2          | 2          | 2          | —          | —          | —          |
| B <sub>5,2</sub>      | 2   | —          | 2          | 2          | —          | —          | —          |
| B <sub>5,3</sub>      | 2   | —          | <b>4</b>   | —          | —          | —          | —          |
| B <sub>6</sub>        | 2   | 2          | 2          | 2          | —          | —          | —          |
| C <sub>3</sub>        | 3   | —          | —          | —          | <b>3</b>   | <b>3</b>   | —          |
| C <sub>4,1</sub>      | 3   | —          | —          | <b>3</b>   | <b>3</b>   | —          | —          |
| C <sub>4,2</sub>      | 3   | —          | —          | <b>3</b>   | <b>3</b>   | —          | —          |
| C <sub>4,3</sub>      | 3   | —          | —          | —          | <b>4</b>   | —          | —          |
| C <sub>5,1</sub>      | 3   | —          | <b>3</b>   | <b>3</b>   | —          | —          | —          |
| C <sub>5,2</sub>      | 3   | —          | <b>3</b>   | <b>3</b>   | —          | —          | —          |
| C <sub>5,3</sub>      | 3   | —          | —          | <b>4</b>   | —          | —          | —          |
| C <sub>5,4</sub>      | 3   | —          | <b>3</b>   | <b>3</b>   | —          | —          | —          |
| C <sub>5,5</sub>      | 3   | —          | —          | <b>4</b>   | —          | —          | —          |
| C <sub>5,6</sub>      | 3   | —          | —          | <b>4</b>   | —          | —          | —          |
| C <sub>5,7</sub>      | 3   | —          | —          | <b>4</b>   | —          | —          | —          |
| C <sub>6</sub>        | 3   | —          | <b>4</b>   | <b>4</b>   | —          | —          | —          |
| $\Sigma^3$ open (cup) | 3   | 3          | —          | —          | —          | —          | —          |

## 2 New exact data for $A_6$ and $A_7$

**The sweep.** Every monomial of a level- $\ell$  denominator factor of  $F_d = \mathcal{N}_d / \prod_r (1 - f_r)$  contains  $x_{l-1}$  exactly once. Hence  $H = G + fH$  is a recurrence along one axis, and multiplying by  $1/(1 - f_r)$  needs one pass over the grid instead of a fixed point. At  $d = 7$  this is 34 passes, not  $\sim 4500$ . Values are computed modulo word-sized primes, reconstructed by CRT, and checked against a prime held back from the reconstruction (`chernpp.bboxes`). The sweep agrees cell by cell with the fixed point of `chernpp.chern`, and packet by packet with `chernpp.crt`.

**Boxes and packets.** Every ballot ordering of  $M$  satisfies  $\beta \leq \beta_{\max}(M)$ , the partial sums of the decreasing ordering. So the level box  $\beta_j \leq (d - j)L$  holds every packet with  $\min M \geq -L$ , and the charge box  $\beta_j \leq jp$  holds every packet with  $\max M \leq p$ . Since  $C(M)$  does not depend on  $\ell$ , one level box gives  $\text{Tp}_{A_d}$  at every  $\ell < L$ .

**Observation 7** (verified).  $\text{Tp}_{A_7}$  for  $\ell \leq 5$  (8033 Chern coefficients) and  $\text{Tp}_{A_6}$  for  $\ell \leq 8$  (12692) have every coefficient positive, and the minimum is 1. Both agree with every published table they overlap. The  $A_7$  box has 49.6M cells and took seven minutes with four primes. The data are in `results/morin_a{6,7}.json`.

### 3 Where positivity is tight

For a zero-sum multiset  $M$  let  $b(M)$  be the number of its *ballot orderings*: distinct orderings whose proper partial sums are nonnegative. Equivalently,  $b(M)$  is the packet sum of the pure chain series  $\prod_j (1 - x_j)^{-1}$ , all of whose coefficients are 1.

**Theorem 8** (plane sector; proved, given two geometric inputs). *If  $\max M \leq 1$  then  $C(M) = b(M)$ . Writing  $M = (1^{d-b}, 1-s_1, \dots, 1-s_b)$  with  $s_i \geq 1$ , this is Kreweras' number of noncrossing partitions of  $[d]$  with block sizes  $s_i$ :*

$$C(M) = b(M) = \frac{d!}{(d-b+1)! \prod_i m_i!},$$

where  $m_i$  is the number of blocks of size  $i$ .

*Proof.* Put  $n = d + 1$ . Four inputs, stated in full in the companion note `report/ballot.pdf` (§2.1), identify  $C(M)$  on this sector with the coefficient described below:

- the curvilinear form of the Bérczi–Szenes incidence formula;
- an elementary extraction of Chern coefficients;
- a multiplicity-one hyperplane lemma;
- Briangon's theorem with the Lehn–Sorger description of the punctual Hilbert scheme of the plane.

Concretely,  $C(M)$  is the coefficient of an  $n$ -cycle  $\sigma$  in  $m_\nu(J_1, \dots, J_n)$ . Here the  $J_k = \sum_{a < k} (a \ k)$  are the Jucys–Murphy elements and  $\nu$  is the partition formed by the entries of  $1 - M$ . Expanding the monomial symmetric function, this coefficient counts pairs  $(e, w)$ . Here  $e = (e_1, \dots, e_n)$  is a rearrangement of  $1 - M$  (with  $e_1 = 0$ , since  $J_1 = 0$ ). And  $w$  is a *minimal monotone factorisation* of  $\sigma$ : a product of  $n - 1$  transpositions  $(a_i \ b_i)$ ,  $a_i < b_i$ , with  $b_1 \leq \dots \leq b_{n-1}$ , exactly  $e_k$  of them having  $b_i = k$ . We show that, for  $\sigma = (x \mapsto x + 1)$ , each  $e$  carries exactly one  $w$  when it satisfies the ballot condition  $D(j) := j - 1 - \sum_{k \leq j} e_k \geq 0$  for all  $j$ , and none otherwise. Reading  $\alpha_k = 1 - e_{k+1}$  identifies such  $e$  with the ballot orderings of  $M$ , which gives  $C(M) = b(M)$ .

*Necessity.* The transpositions of a minimal factorisation of an  $n$ -cycle form a tree on  $[n]$ . Those with  $b_i \leq j$  form a forest on  $[j]$ , so there are at most  $j - 1$  of them.

*Existence and uniqueness,* by induction on  $n$ . Let the transpositions through  $n$  be  $(a_1 \ n) \cdots (a_m \ n)$ , with  $m = e_n$ , and write  $C$  for their product. The rest multiplies to  $F = \sigma C^{-1}$ , a permutation of  $[n-1]$  with its own minimal monotone factorisation. Since  $C^{-1} = (n \ a_1 \ a_2 \ \dots \ a_m)$ , we get  $F(x) = x + 1$  for  $x \notin \{a_i\}$ ,  $F(a_i) = a_{i+1} + 1$ ,  $F(a_m) = 1$ , and  $a_1 = \sigma^{-1}(n) = n - 1$ . As  $FC = \sigma$  is a single cycle and  $C$  is an  $(m + 1)$ -cycle, each cycle of  $F$  contains exactly one  $a_i$ . Following  $x \mapsto x + 1$  from  $a_{i+1} + 1$  we must reach  $a_i$  before any other  $a_j$ , so  $a_1 > a_2 > \dots > a_m$ . Hence  $F$  is the product of the increasing cycles on the intervals  $[1, a_m]$ ,  $[a_m + 1, a_{m-1}]$ ,  $\dots$ ,  $[a_2 + 1, n - 1]$ . Minimality on each interval forces  $D(a_m) = 0$ ,  $D(a_{m-1}) = 1$ ,  $\dots$ , with  $D > i$  strictly inside the  $(i + 1)$ -st interval. So  $a_{m-i}$  is the *last* visit of the walk  $D$  to level  $i$ . Since  $D$  rises by at most 1 per step, these last visits exist and are increasing whenever  $e$  is ballot. The cut points are therefore forced and always available. Commuting transpositions on disjoint intervals merge into a unique monotone order, and the induction closes.

The last equality is the cycle lemma. Append  $-1$  to  $-M$ ; exactly one of the  $n$  cyclic rotations of any arrangement has all proper partial sums nonnegative, so  $b(M) = \frac{1}{n} \binom{n}{d-b+1, m_1, m_2, \dots}$ . The count of monotone factorisations by exponent vector is checked exhaustively for  $n \leq 7$  (the totals are the Catalan numbers 1, 2, 5, 14, 42, 132), and  $C(M) = b(M)$  on every plane-sector packet at  $d \leq 7$  (`chernpp.sectors`, tier 12).  $\square$

So on the plane sector the Morin chamber series behaves exactly like the pure chain series  $\prod_j (1 - x_j)^{-1}$ : every negative  $A_\beta$  there cancels to leave one unit per ballot word. The data say this is the floor everywhere.

**Conjecture 9** (ballot). *For every zero-sum  $M$ ,  $C(M) \geq b(M)$ , with equality exactly when  $\max M \leq 1$ . Equivalently,  $F_d - \prod_j (1 - x_j)^{-1}$  has nonnegative packet sums, vanishing exactly on the plane sector.*

Since the decreasing ordering is always ballot,  $b(M) \geq 1$ . So Conjecture 9 implies Rimányi’s conjecture, and it explains why the least Chern coefficient is 1 and never 0. It holds on every exact coefficient computed, with no violation and no equality off the plane sector. That is 54,390 packets for  $d = 2, \dots, 7$ , including every packet of charge  $\leq 6$  at  $d = 7$  and all of  $\text{Tp}_{A_7}$  for  $\ell \leq 5$  and  $\text{Tp}_{A_6}$  for  $\ell \leq 10$  (results/ballot\_conjecture.json). Off the plane sector the ratio  $C(M)/b(M)$  is at least 2 throughout. Its minimum is  $2, 2, \frac{9}{4}, \frac{12}{5}, \frac{79}{30}, \frac{58}{21}$  for  $d = 2, \dots, 7$ , attained on the family  $(2, 1^{d-4}, -1^{d-2})$  for  $d = 5, 7$ , with a 0 in place of one 1 at  $d = 6$ . A naive refinement, weighting each ballot word by  $\prod_{\alpha_i > 0} \alpha_i!$  (exact for  $(p, -1^p)$ , where  $C = p!$ ), fails already at  $d = 3$ .

A self-contained account, with the evidence table, is in the companion note report/ballot.pdf.

*Remark 10* (what a proof must explain). In the geometric model (stated in report/ballot.pdf, §2.1),  $C(M) = \int_{C_{n,r}} m_\nu(E)$  over the closure of curvilinear length- $n$  subschemes of  $\mathbb{C}^r$ , with the hyperplane lemma lowering  $r$  one step at a time. The plane sector is the case where one can descend to  $r = 2$ , the punctual Hilbert scheme of the plane, where the Jucys–Murphy calculus applies. A packet of charge  $\max M = p$  descends only to  $r = p + 1$ . Conjecture 9 says that the extra dimensions only ever *add*: the integral over the curvilinear locus in  $\mathbb{C}^{p+1}$  is at least the ballot count that the plane would give. This is the precise sense in which charge  $\geq 2$  is where the open problem lives.

Off the plane sector the margin is large. The smallest  $C(M)$  with  $\max M = 2$  is 12, 17, 27 at  $d = 5, 6, 7$ , and the layer minima grow quickly with the charge (Table 2).

**Anatomy of a packet.** Let  $N(M)$  be the negative mass of a packet,  $\sum_{A_\beta < 0} |A_\beta|$  over its ballot orderings, and  $A_{\text{dom}}(M) = A_{\beta_{\max}(M)}$  the coefficient of its decreasing ordering. Table 2 gives, per charge layer, the largest ratio  $N/A_{\text{dom}}$ .

**Conjecture 11** (dominance). *For every  $d$  there is  $\kappa_d < 1$  with  $N(M) \leq \kappa_d A_{\beta_{\max}(M)}$  for every zero-sum  $M$ . In particular  $C(M) \geq (1 - \kappa_d) A_{\beta_{\max}(M)} > 0$ .*

The evidence:  $\kappa_5 = \kappa_6 = \frac{1}{6}$  and  $\kappa_7 = \frac{8}{21}$  on every packet with  $\max M \leq 8, 6, 5$  respectively (exact). The extremal packets are  $(1, 1, 0, -1, -1)$ ,  $(1, 1, 1, 0, -1, -2)$  and  $(2, 1, 1, 0, -1, -1, -2)$ . In every layer the ratio falls by a factor of 3 to 4 per unit of charge, so the conjecture is a statement about small charge. Three remarks keep it honest.

- Individual coefficients are unbounded below:  $\min A_\beta = -3622$  in the  $d = 7$  layer  $p = 5$ . This corrects the report’s earlier “least value  $-2$ ”.
- $\kappa_d$  grows with  $d$ . The conjecture is informative only if it stays below 1, and  $d = 8$  would test that.
- The single term  $A_{\beta_{\max}(M)}$  is the largest in all but 42 of the 3538 packets at  $d = 7$ . A proof would need a lower bound for this one coefficient of  $F_d$  and an upper bound for the negative ones, and no pairing is involved. This is the kind of statement the report’s Assessment §4 asked for.

## 4 The 0-Hecke landscape

On a packet, let  $\pi_i$  act by  $(\pi_i f)(\alpha) = f(\alpha) + f(s_i \alpha)$  if  $\alpha_i > \alpha_{i+1}$ , by  $f(\alpha)$  if  $\alpha_i = \alpha_{i+1}$ , and by 0 otherwise. The  $\pi_i$  satisfy the 0-Hecke relations (tier 11), and  $\Pi_{w_0}$  puts  $C(M)$  on the decreasing ordering. The operators and this criterion are from [4, §3.3]. So  $P_d = \{w : \Pi_w F_d \geq 0\}$  contains  $w_0$  iff Rimányi’s conjecture holds.  $P_d$  is an upper set in the left weak order, and its minimal elements are the minimal cancellation units.

Table 2: Charge layers max  $M = p$  (every packet complete).  $N(M)$  is the negative mass of the packet and  $A_{\text{dom}}$  the coefficient of its decreasing ordering  $\beta_{\text{max}}(M)$ .

| $d$ | $p$ | packets | with $A_\beta < 0$ | $\min C(M)$ | $\min A_\beta$ | $\max N(M)/A_{\text{dom}}$ |
|-----|-----|---------|--------------------|-------------|----------------|----------------------------|
| 5   | 1   | 6       | 1                  | 1           | -1             | 1/6                        |
| 5   | 2   | 23      | 3                  | 12          | -1             | 1/20                       |
| 5   | 3   | 54      | 3                  | 38          | -2             | 1/80                       |
| 5   | 4   | 108     | 3                  | 24          | -3             | 1/258                      |
| 5   | 5   | 185     | 3                  | 336         | -4             | $1.1e - 03$                |
| 5   | 6   | 297     | 3                  | 1296        | -5             | $2.8e - 04$                |
| 5   | 7   | 441     | 5                  | 3696        | -6             | $6.8e - 05$                |
| 5   | 8   | 632     | 5                  | 2880        | -10            | $1.6e - 05$                |
| 6   | 1   | 10      | 2                  | 1           | -1             | 1/6                        |
| 6   | 2   | 47      | 7                  | 17          | -3             | 1/12                       |
| 6   | 3   | 141     | 18                 | 55          | -6             | 3/103                      |
| 6   | 4   | 333     | 35                 | 202         | -20            | 3/296                      |
| 6   | 5   | 674     | 52                 | 120         | -70            | $3.5e - 03$                |
| 6   | 6   | 1226    | 73                 | 2400        | -252           | $1.2e - 03$                |
| 7   | 1   | 14      | 5                  | 1           | -4             | 3/8                        |
| 7   | 2   | 90      | 43                 | 27          | -26            | 8/21                       |
| 7   | 3   | 331     | 134                | 154         | -120           | 43/350                     |
| 7   | 4   | 931     | 339                | 326         | -516           | 9/223                      |
| 7   | 5   | 2172    | 659                | 1284        | -3622          | $1.6e - 02$                |

- $P_5 = S_5 \setminus \langle s_3, s_4 \rangle$ , with minimal elements  $s_1, s_2, s_2s_3, s_2s_3s_4$ . This is unchanged from  $L = 4$  to 5.
- $|P_6| = 643$  with 11 minimal elements, unchanged from  $L = 4$  to 5.  $s_1 \in P_6$ , which is the unpaired-tail and paired inequalities of the  $\tau$  frame holding at  $d = 6$ .
- At  $d = 7$ ,  $s_1 \notin P_7$ , which is the failure of the  $\tau$  frame at  $d = 7$ .  $P_7$  still shrinks with the box ( $|P_7| = 3998, 3352, 2792, 2696$  for  $L = 2, \dots, 5$ ). The minimal parabolic subsets  $J$  with  $w_0(J) \in P_7$  are the same at  $L = 4$  and 5:  $\{1, 2, k\}$  ( $k = 3, \dots, 6$ ) and  $\{1, 3, 5\}$ . At  $d = 5$  they are  $\{1\}$  and  $\{2\}$ , and at  $d = 6$   $\{1\}$  and  $\{2, 5\}$ . The smallest unit of symmetrisation that clears every negative grows from one transposition to three, so no fixed-size pairing survives.

Exclusions from  $P_d$  are proofs, each witnessed by a packet. Memberships hold on the stated boxes only.

## 5 Stabilisation, and what this means for $\mathcal{Q}_8$

**Observation 12** (verified,  $d = 5, 6, 7$ ).  $F_d|_{x_{d-1}=0} = F_{d-1}$ , cell by cell, on the tested boxes. The corresponding statement for numerators,  $[z_d^{e(d)}]\mathcal{Q}_d = (-1)^{e(d)}\mathcal{Q}_{d-1}$  with  $e(d) = \lfloor d^2/4 \rfloor - d + 1$ , was checked directly at  $d = 5$ .

The corank filtration changes the case for  $\mathcal{Q}_8$ . Evidence about generalisations of the conjecture now comes from breadth across coranks, which the registry already supplies. Depth in  $d$  adds one Morin data point, and the published  $\text{Tp}_{A_8}$  and  $\text{Tp}_{A_9}$  at  $\ell = 0$  already lie in  $C_1$ .  $\mathcal{Q}_8$  matters for one question raised here: whether  $\kappa_8 < 1$  in Conjecture 11. The constraints already available for any candidate  $\mathcal{Q}_8$  are its degree (22), its normalisation, the stabilisation  $[z_8^9]\mathcal{Q}_8 = -\mathcal{Q}_7$ , and agreement with Rimányi's  $\text{Tp}_{A_8}$  at  $\ell = 0$ .

Table 3: The 0-Hecke positivity set  $P_d$  restricted to the packets complete in the level box  $\min M \geq -L$ .

| $d$ | $L$ | packets | with $A_\beta < 0$ | $ P_d $ | minimal elements |
|-----|-----|---------|--------------------|---------|------------------|
| 5   | 5   | 377     | 12                 | 114     | 4                |
| 6   | 4   | 532     | 39                 | 643     | 11               |
| 6   | 5   | 1206    | 76                 | 643     | 11               |
| 7   | 2   | 105     | 39                 | 3998    | 23               |
| 7   | 3   | 436     | 141                | 3352    | 49               |
| 7   | 4   | 1367    | 374                | 2792    | 30               |
| 7   | 5   | 3539    | 845                | 2696    | 33               |

## Reproducing

- `tools/morin_tables.py 7 6 results/morin_a7.json` (the  $A_7$  tables).
- `tools/packet_anatomy.py`, `tools/hecke_landscape.py`, and `tools/corank_survey.py` for  $\rho$ .
- `tools/render_d7_tables.py report/tables` regenerates every table here.

Tiers 10–14 of the test suite pin each module.

## References

- [1] L. Bérczi, A. Szenes, Thom polynomials of Morin singularities, *Ann. of Math.* 175 (2012) 567–629.
- [2] R. Rimányi, Thom polynomials, symmetries and incidences of singularities, *Invent. Math.* 143 (2001) 499–521.
- [3] P. Pragacz, A. Weber, Positivity of Schur function expansions of Thom polynomials, *Fund. Math.* 195 (2007) 85–100.
- [4] Rimányi positivity: findings summary (29 July 2026), unrefereed, `papers/rimanyi_positivity_findings_summary.pdf`.